

VAXPLAN

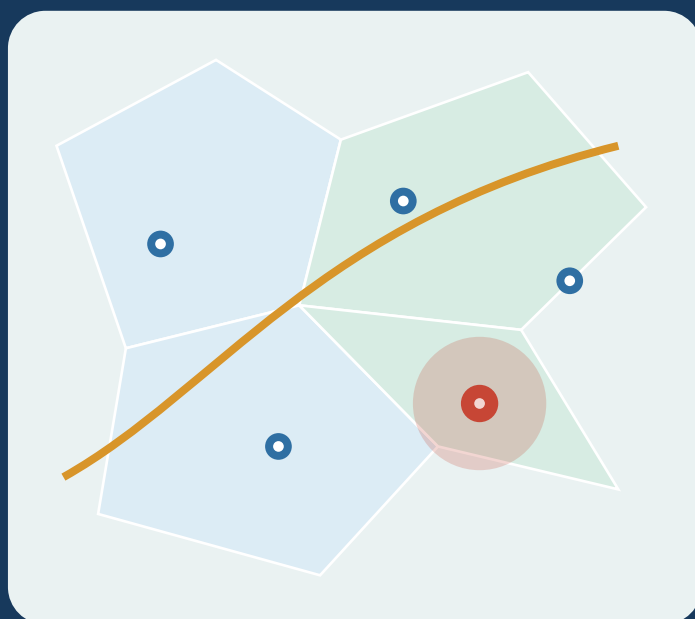
PARTNERSHIP & IMPLEMENTATION

VaxPlan Partnership & Implementation Brief

2026–2030

GIS-Enabled Immunisation Microplanning, Operational Intelligence and Decision Support

**Reach every child.
Plan every session.**



Executive Summary

VaxPlan is a GIS-enabled immunisation planning, operational intelligence and decision-support platform. It helps programmes translate population, settlement, facility, accessibility and performance data into practical microplans, service-delivery choices and monitorable vaccination sessions.

The opportunity

Planning information is often distributed across maps, spreadsheets and programme systems. VaxPlan provides a structured operational layer without displacing those systems.

Why GIS

Location changes the decision: who is underserved, which facility is responsible, where outreach is feasible and which route or delivery strategy fits.

How it integrates

API-ready workflows can consume approved data from DHIS2, registries, LMIS, facility registries, census, field tools and geospatial sources.

The partnership

Co-design a bounded proof of concept, evaluate operational value and readiness, and scale only after agreed evidence and governance gates.

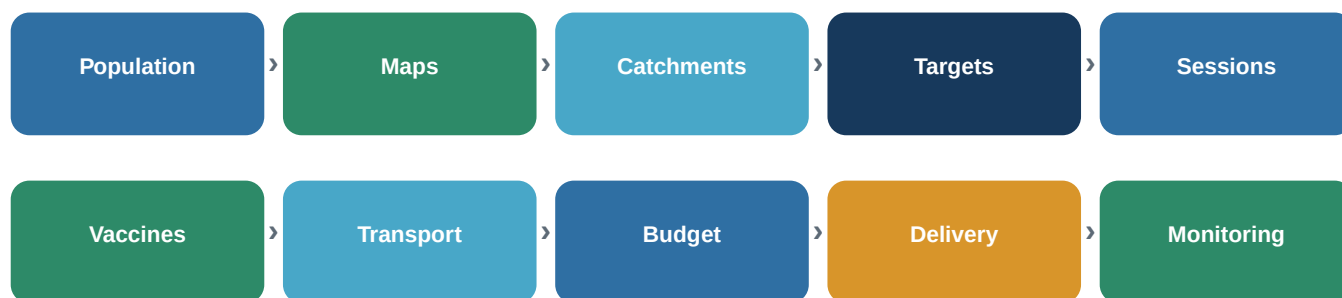
Proposed collaboration pathway



VaxPlan is designed to complement national architecture—not replace DHIS2, electronic immunisation registries, logistics systems or field data-collection tools.

The Planning Challenge

Immunisation teams already hold valuable data and expertise. The practical challenge is connecting these assets into a repeatable planning cycle that reaches from national priorities to the last settlement.



Common friction points

Fragmented inputs

Approved estimates, geographic data and delivery records may live in different systems.

Repeated preparation

Plans are frequently reconstructed instead of reused, reviewed and progressively improved.

Limited spatial translation

Data describe performance, but do not always reveal which settlement or access barrier needs action.

Weak feedback loops

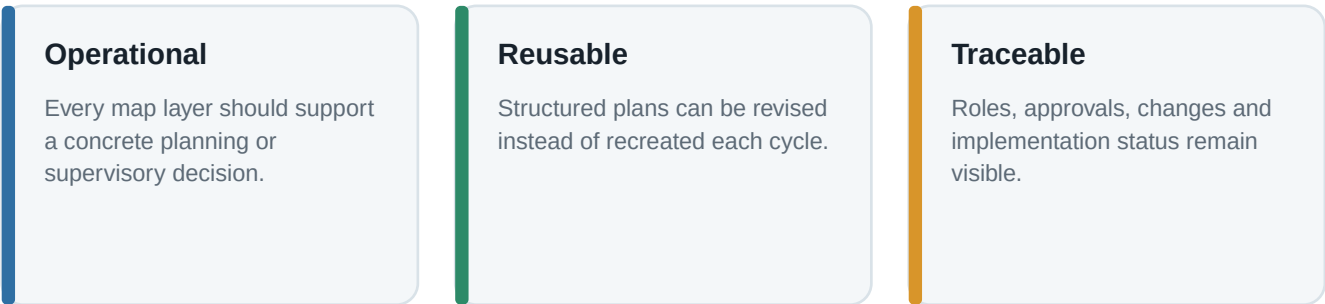
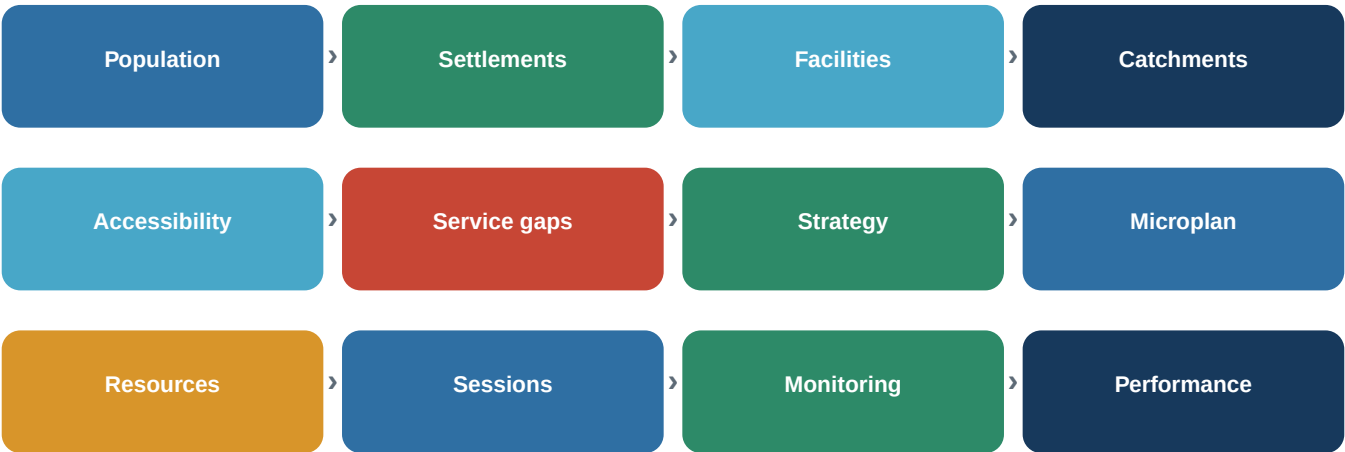
Completed sessions and local learning do not consistently update the next planning cycle.

The opportunity is not another parallel reporting system. It is a coherent planning and decision layer that makes existing data operational.

The VaxPlan Approach

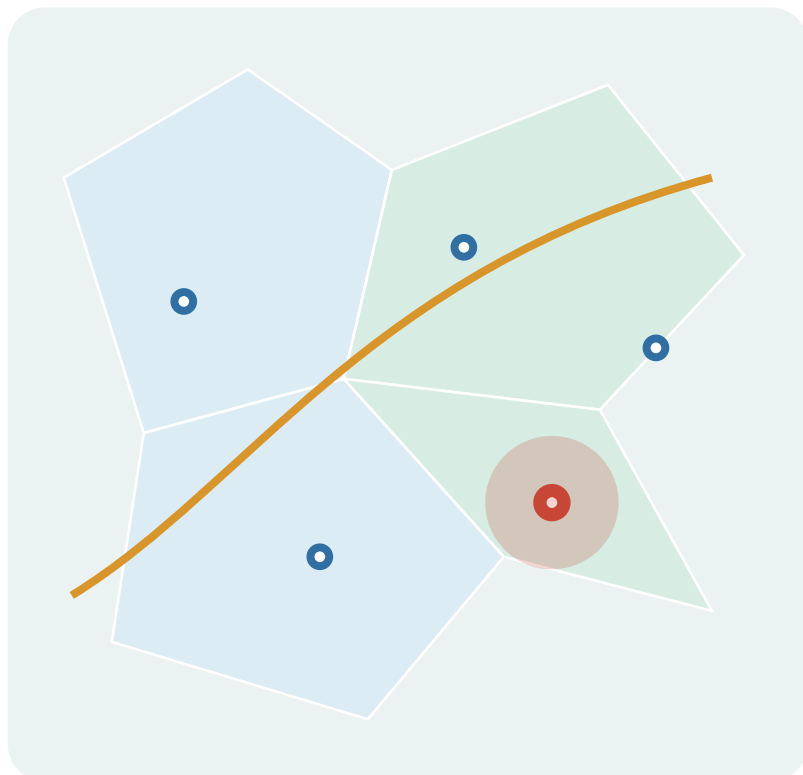


From population intelligence to accountable delivery



The result is a shared operational picture for facility, district and national teams—adapted to programme rules and country governance.

Geospatial Intelligence



A map is useful when it changes a decision

VaxPlan connects spatial context to catchment definition, service strategy, resource planning and follow-up. It can combine approved facility and settlement coordinates with population surfaces, boundaries and access constraints.

Facilities & communities

See responsibility, proximity and unserved areas.

Access & travel

Use distance, routes and context to assess feasibility.

Catchments & gaps

Compare expected service extent with planned delivery.

Population overlays

Prioritise places using approved denominators and spatial estimates.

Operational action

Open or revise a microplan, session or follow-up action from the geographic insight.

Illustrative visual only. Operational decisions must use country-approved data, thresholds and programme rules.

Zero-Dose and Equity

VaxPlan can surface geographic patterns associated with missed services: remote settlements, catchment gaps, high population clusters, difficult travel and planned-session absence. These signals help programmes prioritise investigation and action.

Geographic zero-dose risk

A planning inference based on place, population, access and service-delivery evidence. It points teams to areas that may be underserved.

Confirmed individual status

Requires appropriate individual-level registry or verified programme data. VaxPlan should not claim individual zero-dose status without this evidence.

Equity-oriented decision cycle



- Identify settlements and populations outside expected service reach.
- Examine accessibility, gender, demand, insecurity and other locally relevant barriers.
- Select fixed, outreach or mobile strategies using national policy and field judgement.
- Monitor planned versus completed sessions and update the evidence base.

Digital Microplanning

Structured workflows convert programme guidance into a transparent, reviewable operational plan while preserving space for local judgement and comments.

1**Area & targets**

Catchments, communities, denominators and delivery strategy

2**Sessions & teams**

Calendar, session type, workload and staffing

3**Supplies & access**

Vaccines, consumables, cold chain, transport and budget

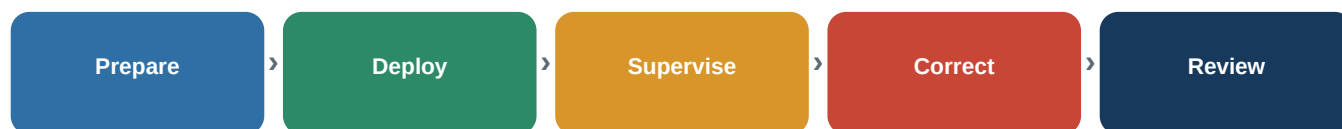
4**Review & learn**

Supervision, approvals, monitoring and revision

Role-based workflow gates and an approval audit trail support programme governance.
Requirements should be configured to the national process, not hard-coded as universal rules.

Campaigns and Monitoring

Campaign operations require a faster cadence than routine planning while still benefiting from shared geographic and resource foundations.



Readiness

Operational areas, teams, supplies, training and deployment readiness.

Implementation

Daily activity, missed places, emerging constraints and corrective action.

Supervision

Structured checks, escalation, geographic oversight and accountability.

Learning

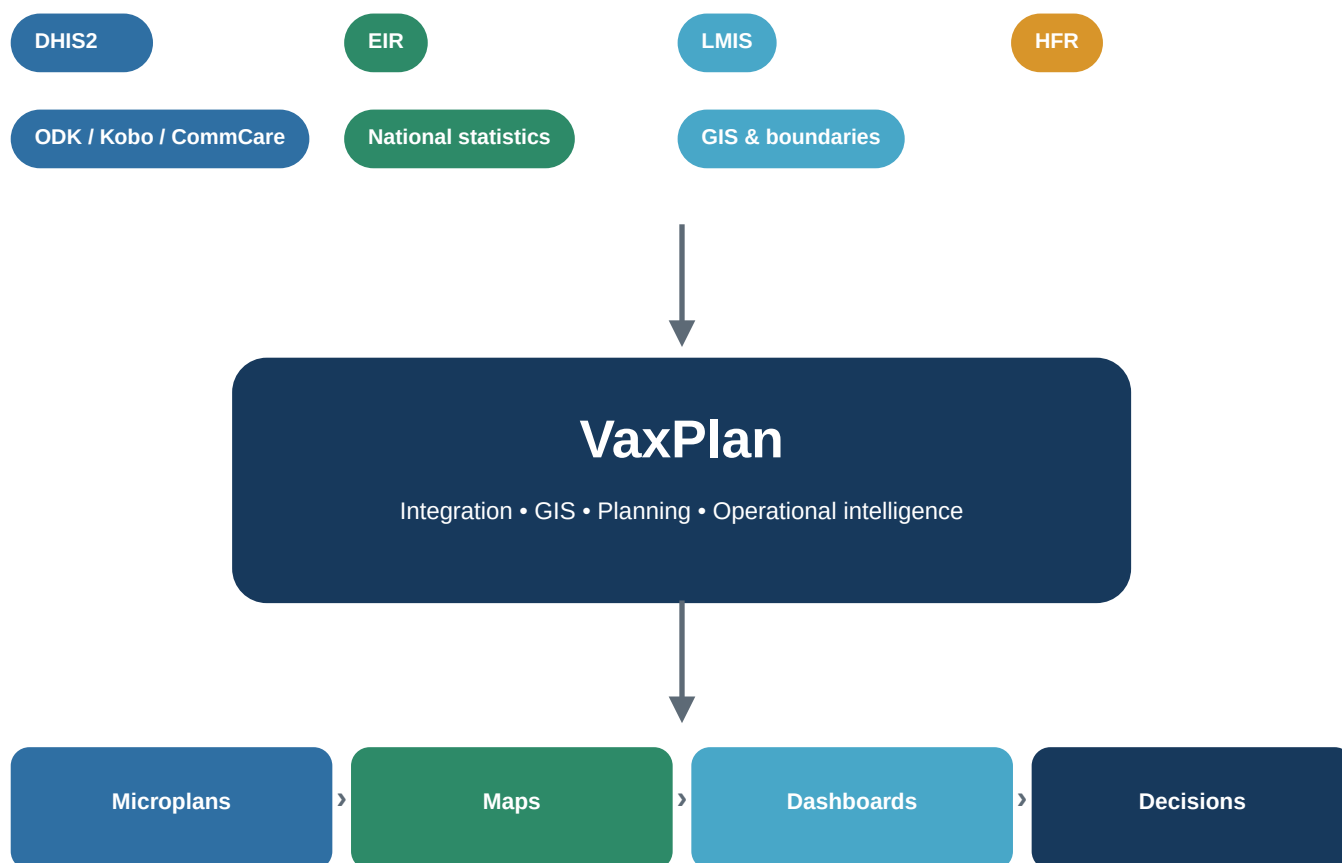
Post-campaign review, performance patterns and reusable improvements.

Separate programme semantics

Routine and supplementary immunisation activities require distinct plan types, schedules, targets, session records and rollups. Shared infrastructure should never blur their totals or accountability.

Interoperability

VaxPlan is an integration and intelligence layer. It is designed to complement national systems and reuse authoritative data through governed exchanges.

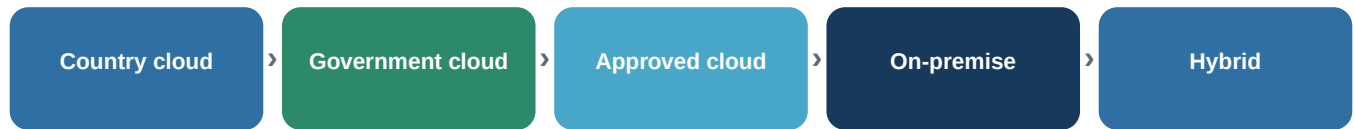


Integration principles

- Country-owned identifiers and master data.
- Documented, versioned APIs and standards-based exchange where feasible.
- Least-privilege access, auditability and data minimisation.
- No parallel reporting requirement unless explicitly agreed for a bounded pilot.

Deployment Architecture

Deployment should follow country policy, data classification, connectivity and operational ownership—not a one-size-fits-all hosting assumption.



User experience

Responsive web application; installable/offline patterns where configured

Application services

APIs, workflow logic, validation, reporting and integrations

Data services

PostgreSQL/PostGIS, governed files, backups and retention

Security & operations

RBAC, encryption, audit logs, monitoring, recovery and incident response

A production deployment requires a signed security and operations design: identity, environments, backups, recovery objectives, observability, patching, support and data-processing responsibilities.

Partnership Models

Demonstrate

1–2 sessions

Validate use cases, workflow fit and stakeholder interest.

Purpose confirmed

Pilot

3–6 months

Configure, train, deploy and evaluate a bounded implementation.

Scope agreed

Scale

Phased expansion

Extend proven workflows, integrations and support capacity.

Outputs documented

Nationalise

Institutional ownership

Localise, govern, operate and continuously improve nationally.

Decision gate

Partnership contributions

Country leadership and programme expertise • authoritative data and policy • implementation and integration capacity • independent evaluation • catalytic financing • long-term operational ownership

Example Proof of Concept

3

districts

20–30

health facilities

3–6

months

1

Discovery

Confirm priorities, users and governance

2

Configuration

Localise workflows and reference data

3

Data preparation

Validate facility, settlement and denominator sources

4

Training

Prepare administrators, planners and reviewers

5

Deployment

Launch bounded operational use

6

Support

Resolve issues and coach teams

7

Evaluation

Measure usability, operations and outcomes

8

Scale decision

Agree stop, adapt or expand

Illustrative scope only. Final design and budget are co-developed after discovery and data-readiness review.

Monitoring and Evaluation Framework

A pilot should begin with an agreed theory of change, baseline, indicator definitions, data sources and decision gates.

Adoption

Facilities with completed microplans; active users; user satisfaction

Efficiency

Time to develop or update a microplan; reuse of approved reference data

Geographic equity

Previously unmapped settlements; population spatially allocated; service gaps identified

Delivery

Planned versus completed outreach sessions; cancelled or rescheduled sessions

Data quality

Synchronisation success; validation exceptions; denominator confidence

Decision use

Programme decisions informed by GIS outputs; corrective actions closed

Reliability

Availability, incident resolution and backup/recovery tests

Institutionalisation

Trainers prepared; governance routines; recurrent financing and support ownership

No impact claim should be made without an appropriate evaluation design and verified programme data.

Sustainability and Ownership

Sustainability is designed through governance, people, standards and financing—not added at the end of a pilot.

Government ownership

Named product owner, steering mechanism and decision rights.

Local capacity

Master trainers, administrators, support teams and documented procedures.

Interoperability

Reuse authoritative registries and standards; avoid avoidable duplication.

Progressive handover

Milestones for configuration, operations, support and roadmap ownership.

Total cost of ownership

Hosting, connectivity, support, integrations, devices, training and change management.

Reusable configuration

Country rules and workflows maintained as governed assets.

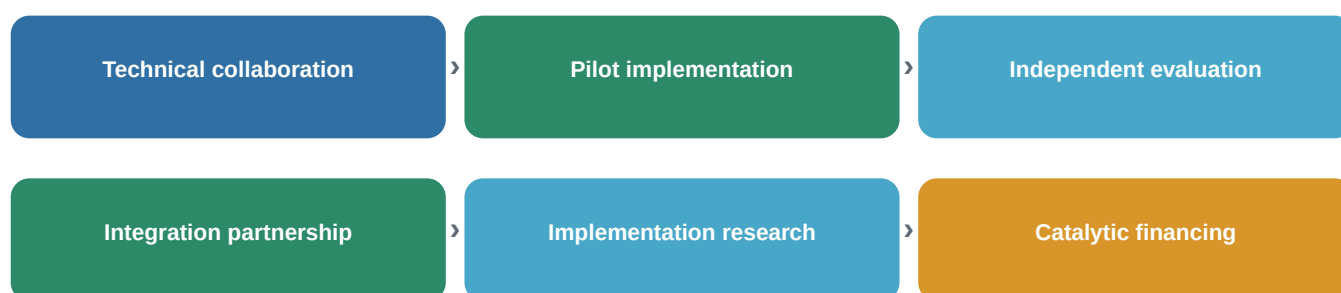
A responsible scale decision

Proceed when the programme has evidence of workflow value, data readiness, user adoption, integration feasibility, security acceptability and an affordable operating model.

Partnership Invitation

Build the Next Generation of Immunisation Microplanning With Us

VaxPlan invites Ministries of Health, EPI programmes, WHO, UNICEF, Gavi, Africa CDC, research institutions, implementation partners and donors to explore a country-led demonstration or proof of concept.



Start the conversation

Request a demonstration

<https://vaxplan.org/demo>

Explore partnership

<https://vaxplan.org/partners>

Selected references

World Health Organization. Reaching Every District (RED): a guide to increasing coverage and equity in all communities in the African Region (2017).

World Health Organization. Digital Implementation Investment Guide: Integrating Digital Interventions into Health Programmes (2020).

World Health Organization. SMART Guidelines: standards-based, machine-readable, adaptive, requirements-based and testable guidance.

UNICEF. Digital health and information systems: digitally enabled health systems drive results for children.

Gavi, the Vaccine Alliance. Zero-dose children and missed communities; equity goal and operational definitions.

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